Fixed Point Theorems in Banach Spaces

Banach Contraction Mapping Theorem

$\|T(x) - T(y)\| \leq c\|x - y\|$

$c < 1 \Rightarrow$ *unique fixed point*

Brouwer Fixed Point Theorem

$T : B \to B$ continuous

B compact convex

Schauder Fixed Point Theorem

T continuous, compact

Self-map on compact convex

Requirements for Fixed Points:

Completeness: $(X, \|\cdot\|)$ is Banach space

Continuity: $T : X \to X$ is continuous

Additional: Contraction, compactness, or convexity

Key Applications:

Solvability of integral equations, existence of solutions to PDEs, iterative algorithms